MA1521 CALCULUS FOR COMPUTING

National University of Singapore

Exercise Set F (Final Exam Revision)

Name: _____

Date: _____

Partial Derivatives and Multiple Integrals

Thomas' Calculus: Early Transcendentals, 15/e • Sections 11.3–11.5, Chapter 13, and Chapter 14 complete

MIT 18.02 Multivariable Calculus • Lectures 1–18 and 25–26

HOW TO USE THIS SET

The sample final paper has 28 short-answer questions: three on single-variable calculus, nine on vectors and partial derivatives, and sixteen on multiple integrals. This set is weighted the same way. Parts A to C revise the material already examined at the midterm, in the form the final asks it; Parts D to F cover Chapter 14 in full; Part G is a complete 28-question mock paper to be sat in one sitting of two hours.

Every problem here is new: none is copied from the textbook, the lesson notes, or the sample paper, although each is modelled on a question type that appears there. Every problem is followed by a ruled panel sized to the work it should take. Answers to all problems begin on the last pages; consult them only after a complete attempt.

Distribution of Problems

Part	Topic	Thomas sections	Problems
A	Single-Variable Warm-Up	4.5, 5.4, 8.8	1–3
B	Lines, Planes, and Partial Derivatives	11.3–11.5, 13.3–13.6	4–9
C	Extreme Values and Lagrange Multipliers	13.7–13.8	10–14
D	Double Integrals in Rectangular Coordinates	14.1–14.3	15–22
E	Polar Coordinates and Substitutions	14.4, 14.8	23–33
F	Triple Integrals	14.5–14.7	34–45
G	Mock Final Examination (2 hours)	all of the above	46–73

Results You Are Expected to Know Cold

THEOREM — Fubini's Theorem and Reversing the Order of Integration

If f is continuous on $R:\ a \le x \le b,\ g_1(x) \le y \le g_2(x)$, then

$$\iint_R f(x,y)\,dA = \int_a^b \int_{g_1(x)}^{g_2(x)} f(x,y)\,dy\,dx.$$

The outer limits are always constants. To reverse the order, sketch R, describe it again as $c \le y \le d,\ h_1(y) \le x \le h_2(y)$, and rewrite the limits; never simply swap the integral signs. On a rectangle, $\iint g(x)h(y)\,dA = \left(\int g\,dx\right)\left(\int h\,dy\right)$.

THEOREM — Substitutions in Multiple Integrals

If $x = g(u,v)$, $y = h(u,v)$ maps a region G of the uv-plane one-to-one onto R, then

$$\iint_R f(x,y)\,dx\,dy = \iint_G f\big(g(u,v),h(u,v)\big)\left|\frac{\partial(x,y)}{\partial(u,v)}\right|du\,dv, \qquad \frac{\partial(x,y)}{\partial(u,v)} = \begin{vmatrix} x_u & x_v \\ y_u & y_v \end{vmatrix} = \frac{1}{\partial(u,v)/\partial(x,y)}.$$

Polar: $dA = r\,dr\,d\theta$ Cylindrical: $dV = r\,dz\,dr\,d\theta$ Spherical: $dV = \rho^2 \sin\phi\,d\rho\,d\phi\,d\theta$

In spherical coordinates $x = \rho\sin\phi\cos\theta$, $y = \rho\sin\phi\sin\theta$, $z = \rho\cos\phi$, with $\rho \ge 0$ and $0 \le \phi \le \pi$.

PART A
Single-Variable Warm-Up

3 problems

Thomas 4.5, 5.4, 8.8 • sample final Questions 1–3

Improper Integrals

1. Improper integrals. Each integral below is improper. Say why, and evaluate it.

(a) $\displaystyle\int_0^1 x^2(\ln x)^2\,dx$

(b) $\displaystyle\int_0^1 \frac{\ln x}{\sqrt{x}}\,dx$

(c) $\displaystyle\int_1^\infty \frac{\ln x}{x^3}\,dx$

(d) $\displaystyle\int_0^\infty x^2 e^{-3x}\,dx$

Indeterminate Forms and L'Hôpital's Rule

2. Limits. Evaluate each limit.

(a) $\displaystyle\lim_{x\to 3} \frac{(x-3)(\ln x - \ln 3)}{x^2 - 7x + 12 + 3\ln x - 3\ln 3}$

(b) $\displaystyle\lim_{x\to 0} \frac{e^{3x} - 1 - 3x}{1 - \cos 2x}$

(c) $\displaystyle\lim_{x\to \infty} \left(1 + \frac{2}{x}\right)^{3x}$

The Fundamental Theorem of Calculus

3. Differentiating an integral.

(a) Let $f(x) = \displaystyle\int_{2x-1}^{x^3} \frac{1}{t^2+4}\,dt$. Find $f'(2)$.

(b) Let $g(x) = \displaystyle\int_{x}^{x^2} \sqrt{1+t^3}\,dt$. Find $g'(2)$.

(c) Let $F(x) = \displaystyle\int_{0}^{x^2} (t-4)e^{-t}\,dt$ for $x > 0$. At what value of x does F have a local minimum?

PART B
Lines, Planes, and Partial Derivatives

6 problems

Thomas 11.3–11.5, 13.3–13.6 • sample final Questions 4–9

Lines and Planes in Space

4. Three short questions.

(a) Let $P = (1, -2, 3)$ and $Q = (4, 0, 1)$. Find the point at which the line through P and Q meets the plane $3x - y + 2z = 5$.

(b) The angle between the vectors $\mathbf{i} - 2\mathbf{j} + \mathbf{k}$ and $-2\mathbf{i} + \mathbf{j} + \mathbf{k}$ is $a\pi$ with $0 \le a \le 1$. Find a.

(c) Find an equation of the plane through $(1, 1, 0)$, $(2, 0, 3)$, and $(0, 2, 1)$, and the distance from the point $(3, 4, 5)$ to that plane.

Tangent Planes

5. Tangent planes to level surfaces.

(a) Let S be the surface $x^2 + 4y^2 - z = 5$. Find the point on S at which the tangent plane to S is parallel to the plane $4x - 8y - 2z = 1$.

(b) Find an equation of the tangent plane to the surface $xyz + z^3 = 12$ at the point $(1, 2, 2)$.

Recovering a Function from Its Partial Derivatives

6.

(a) Find the function $f(x, y)$ such that

$$\frac{\partial f}{\partial x} = 2xe^y - y\sin x, \qquad \frac{\partial f}{\partial y} = x^2e^y + \cos x + 2y, \qquad f(0,0) = 3,$$

and hence find $f(\pi, 1)$.

(b) For what value of the constant k does there exist a function $g(x, y)$ with

$$\frac{\partial g}{\partial x} = kxy^2 + 3y, \qquad \frac{\partial g}{\partial y} = 4x^2y + 3x + 1\,?$$

For that value of k, find all such functions g.

The Chain Rule

7. Let $u = 2xy$, let $v = v(x, y)$ be a function such that $\dfrac{\partial v}{\partial x} = 4x$ and $\dfrac{\partial v}{\partial y} = -y$, let $w = x^2 - y$, and let

$$z = u^2 - 3v + uw.$$

Determine $\dfrac{\partial z}{\partial x}$ as a function of x and y.

Directional Derivatives and Rates of Change

8. Consider the function $f(x, y) = xy^2 - x^3$.

(a) Find the directional derivative of f at the point $(1, 2)$ in the direction of the vector $5\mathbf{i} - 12\mathbf{j}$.

(b) What is the largest value the directional derivative of f at $(1, 2)$ can take, over all directions?

9. Let $T(x, y, z)$, measured in degrees Celsius, be the temperature at the point (x, y, z) of a server room, where x, y, z are measured in metres. It is given that

$$\nabla T(x, y, z) = \left\langle\, y^2 - z,\ 2xy,\ 4 - x \,\right\rangle.$$

A sensor drone flies through the room so that its position at time t seconds is $\mathbf{r}(t) = \langle t^2,\ 2t - 1,\ t^3 \rangle$. Find the rate of change, in degrees Celsius per second, of the temperature recorded by the drone at time $t = 1$.

PART C
Extreme Values and Lagrange Multipliers

5 problems

Thomas 13.7–13.8 • sample final Questions 10–12

Critical Points and the Second Derivative Test

10. Find all the critical points of

$$f(x, y) = x^2y - 2xy + 2y^2 - 15y,$$

and classify each as a local maximum point, a local minimum point, or a saddle point. Give the value of f at each local extremum.

11. Determine all the saddle points and all the local extrema of

$$f(x, y) = x^3 - 12xy + 8y^3.$$

Absolute Extrema on a Closed Disk

12. Let $f(x, y) = x^2 + 2y^2 - x$.

(a) Find the critical point of f and its type.

(b) Write down the system of three equations in x, y, and λ that the method of Lagrange multipliers gives for the extreme values of f on the circle $x^2 + y^2 = 1$.

(c) Find the maximum and minimum values of f on the closed disk $x^2 + y^2 \leq 1$.

The Method of Lagrange Multipliers

13. Constrained extrema.

(a) Find the maximum and minimum values of $f(x, y) = xy^2$ on the circle $x^2 + y^2 = 3$.

(b) Find the point on the plane $2x - y + 2z = 18$ closest to the origin, and its distance from the origin.

14. Three positive numbers x, y, z have sum 12. Find the largest possible value of the product xy^2z^3, and the values of x, y, z at which it is attained.

PART D
Double Integrals in Rectangular Coordinates

8 problems

Thomas 14.1–14.3 • sample final Questions 13–17

Double Integrals over Rectangles

15. Rectangles. Evaluate each double integral.

(a) $\displaystyle\iint_R \left(4x^3 + 6xy^2\right) dA, \quad R:\ 0 \le x \le 1,\ 0 \le y \le 2$

(b) $\displaystyle\iint_R x\cos(xy)\, dA, \quad R:\ 0 \le x \le \pi/2,\ 0 \le y \le 1$

(c) $\displaystyle\iint_R \frac{y}{1+xy}\, dA, \quad R:\ 0 \le x \le 1,\ 0 \le y \le 1$

In parts (b) and (c) one order of integration is much easier than the other.

16. Integrals of products.

(a) Let f be a continuous function with $\int_0^8 f(t)\,dt = 12$, and let R be the rectangle bounded by the lines $x = 0$, $x = 4$, $y = 0$, $y = 2$. Find the value of $\iint_R f(2x)\,f(4y)\,dA$.

(b) Let g be a continuous function with $\int_0^1 g(t)\,dt = 5$, and let S be the unit square $0 \le x \le 1$, $0 \le y \le 1$. Find the value of $\iint_S \big(g(x) + g(y)\big)\,dA$.

Double Integrals over General Regions

17. Consider the triangular region R in the xy-plane with vertices $(1, 1)$, $(5, 1)$, and $(3, 5)$. Compute $\iint_R y\,dA$.

18. Let R be the region in the xy-plane bounded by the parabola $y = x^2$ and the line $y = 2x + 3$.

(a) Find the area of R.

(b) Find $\iint_R x\,dA$, and deduce the x-coordinate of the centroid of R.

19. Let R be the region in the first quadrant bounded by the curve $y = \sqrt{x}$, the line $x + y = 2$, and the x-axis. Sketch R, and evaluate $\iint_R y\,dA$ using a *single* iterated integral.

Reversing the Order of Integration

20. Reversing the order. Sketch the region of integration and evaluate each iterated integral by reversing the order of integration.

(a) $\displaystyle\int_0^1 \int_{\sqrt{x}}^1 \sqrt{1+y^3}\, dy\, dx$

(b) $\displaystyle\int_0^4 \int_{y/2}^2 e^{x^2}\, dx\, dy$

(c) $\displaystyle\int_0^{\pi/2} \int_x^{\pi/2} \frac{\cos y}{y}\, dy\, dx$

(d) $\displaystyle\int_0^8 \int_{x^{1/3}}^2 \frac{1}{y^4+1}\, dy\, dx$

21. The sum

$$\int_0^1 \int_0^x xy\,dy\,dx + \int_1^2 \int_0^{2-x} xy\,dy\,dx$$

is a double integral of xy over a single region R. Sketch R, write the sum as one iterated integral with the order $dx\,dy$, and evaluate it.

Volume

22. Find the volume of the solid that lies below the plane $z = 2x + y + 1$ and above the region in the xy-plane bounded by the parabola $y = x^2$ and the line $y = 1$.

PART E
Polar Coordinates and Substitutions

11 problems

Thomas 14.4, 14.8 • sample final Questions 18–22

Double Integrals in Polar Form

23. Disks, annuli, and sectors. Evaluate each double integral.

(a) $\iint_R e^{-(x^2+y^2)}\,dA$, where R is the closed disk of radius 2 centred at the origin.

(b) $\iint_R \left(x^2+y^2\right)^{3/2}\,dA$, where R is the part of the annulus $1 \le x^2+y^2 \le 4$ that lies in the upper half-plane $y \ge 0$.

(c) $\iint_R y\,dA$, where R is the region in the first quadrant bounded by the circles $x^2+y^2=1$ and $x^2+y^2=9$ and the lines $y=x$ and $y=\sqrt{3}\,x$.

24. Circles through the origin.

(a) Let R be the region in the first quadrant bounded by the curve $y = \sqrt{4x - x^2}$ and the x-axis. Find $\displaystyle\iint_R \frac{1}{\sqrt{x^2 + y^2}}\, dA$.

(b) Let R be the closed disk $x^2 + y^2 \leq 2y$. Find $\displaystyle\iint_R y\, dA$ using polar coordinates, and explain how the answer could have been predicted without integrating.

25.

(a) Compute $\dfrac{1}{\pi}\iint_R \left(x^4 + 2x^2y^2 + y^4 - 2x^2 - 2y^2 + 4\right) dA$, where R is the closed disk of radius $\sqrt{2}$ centred at the origin.

(b) Evaluate $\displaystyle\int_0^2 \int_y^{\sqrt{8-y^2}} \frac{1}{1 + x^2 + y^2}\, dx\, dy$ by changing to polar coordinates.

26. Find the area enclosed by one leaf of the rose $r = 2\cos 3\theta$.

27. Let $I = \displaystyle\int_0^{\infty} e^{-4x^2}\,dx$. By writing I^2 as a double integral over the first quadrant and changing to polar coordinates, find the exact value of I.

Substitutions in Double Integrals

28. Let D be the parallelogram in the xy-plane bounded by the lines

$$x - 2y = 0, \qquad x - 2y = 4, \qquad 2x + y = 1, \qquad 2x + y = 6.$$

Compute $\displaystyle\iint_D (x - 2y)\, e^{2x+y}\,dx\,dy$, giving the answer in terms of integer powers of e.

29. Let R be the region in the first quadrant of the xy-plane bounded by the hyperbolas $xy = 1$ and $xy = 2$ and the lines $y = x$ and $y = 4x$. Use the change of variables $u = xy$, $v = y/x$ to compute $\iint_R y^2 \, dA$.

30. Let R be the region in the first quadrant of the xy-plane bounded by the curves $x^2 - y^2 = 1$, $x^2 - y^2 = 3$, $xy = 1$, and $xy = 4$. Compute $\iint_R \left(x^4 - y^4\right) dA$.

Suggestion. The Jacobian $\partial(u, v)/\partial(x, y)$ is much easier to compute than $\partial(x, y)/\partial(u, v)$.

31. Consider the double integral

$$\iint_R \frac{x^2}{y} \cos\left(y^2\right) dA,$$

where R is the region in the first quadrant of the xy-plane bounded by the lines $y = 1$, $y = 2$, $y = x$, and $y = 3x$. Use the change of variables $u = y^2$ and $v = x/y$ to rewrite the integral as an iterated integral in the variables u, v with integration order $du\, dv$.

[You do not have to compute the integral.]

32. Let R be the region enclosed by the ellipse $\dfrac{x^2}{9} + \dfrac{y^2}{4} = 1$. Evaluate

$$\iint_R \left(\frac{x^2}{9} + \frac{y^2}{4}\right) dA.$$

33. Let R be the triangle with vertices $(0,0)$, $(2,0)$, and $(0,2)$. Use the change of variables $u = y - x$, $v = y + x$ to evaluate

$$\iint_R e^{(y-x)/(y+x)}\, dA.$$

PART F
Triple Integrals

12 problems

Thomas 14.5–14.7 • sample final Questions 23–28

Triple Integrals in Rectangular Coordinates

34. Iterated integrals. Evaluate.

(a) $\displaystyle\int_0^2\int_0^1\int_0^3 \left(x^2y + 2z\right)\,dz\,dy\,dx$

(b) $\displaystyle\int_0^1\int_0^{2x}\int_0^{x+y} 6z\,dz\,dy\,dx$

(c) $\displaystyle\int_{-1}^1\int_{-2}^2\int_{-3}^3 \left(x^3y^2 + z\sin y + 5\right)\,dz\,dy\,dx$

Part (c) should take one line.

35. Let D be the tetrahedron in the first octant bounded by the coordinate planes and the plane $x + 2y + z = 2$.

(a) Find the volume of D.

(b) Evaluate $\iiint_D y \, dV$.

(c) Find the centre of mass of D, assuming constant density.

Triple Integrals in Cylindrical Coordinates

36. Find the volume of the solid bounded below by the paraboloid $z = x^2 + y^2$ and above by the plane $z = 2y$.

37. Let R be the solid region bounded by the cylinders $x^2 + y^2 = 1$ and $x^2 + y^2 = 4$ and the planes $z = 0$ and $z = 2$. Compute

$$\iiint_R z\left(x^2 + y^2\right)^{3/2}\, dx\, dy\, dz.$$

38. Consider the solid region E bounded by the cylinder $x^2 + y^2 = 4$, the horizontal plane $z = 1$, and the plane $z = 5 - y$. Evaluate

$$\frac{1}{\pi}\iiint_E \left(x^2 + y^2\right) dV.$$

39. Let E be the solid consisting of the points (x, y, z) with $\sqrt{x^2 + y^2} \leq z \leq 3$. Evaluate $\iiint_E z \, dV$.

40. Evaluate

$$\int_{-2}^{2} \int_{-\sqrt{4-x^2}}^{\sqrt{4-x^2}} \int_{\sqrt{x^2+y^2}}^{2} \left(x^2 + y^2\right) dz \, dy \, dx$$

by converting to cylindrical coordinates.

Triple Integrals in Spherical Coordinates

41. Balls and shells. Evaluate.

(a) $\iiint_E \sqrt{x^2+y^2+z^2}\,dV$, where E is the set of points with $1 \le x^2+y^2+z^2 \le 4$ and $z \ge 0$.

(b) $\iiint_E z\,dV$, where E is the upper half of the ball of radius 2 centred at the origin.

(c) $\iiint_E x^2\,dV$, where E is the unit ball $x^2+y^2+z^2 \le 1$.

42. Find the volume of the solid that lies above the cone $z = \sqrt{3\left(x^2+y^2\right)}$ and inside the sphere $x^2+y^2+z^2 = 4$.

43.

(a) Let E be the solid consisting of the points (x, y, z) with $\sqrt{3\left(x^2+y^2\right)} \leq z \leq 3$. Rewrite

$$\iiint_E \frac{1}{1+x^2+y^2+z^2}\, dV$$

as an iterated integral in spherical coordinates with integration order $d\rho\, d\phi\, d\theta$. *[You do not have to compute the integral.]*

(b) Let B be the ball $x^2+y^2+z^2 \leq 2z$. Evaluate $\iiint_B \sqrt{x^2+y^2+z^2}\, dV$.

Masses, Centres of Mass, and Other Substitutions

44. A solid is bounded below by the paraboloid $z = x^2 + y^2$ and above by the plane $z = 4$. Its density at the point (x, y, z) is $\delta = z$. Find the mass of the solid and the z-coordinate of its centre of mass.

45. Let E be the solid ellipsoid $\dfrac{x^2}{4} + \dfrac{y^2}{9} + z^2 \leq 1$. Use the substitution $x = 2u$, $y = 3v$, $z = w$ to evaluate $\iiint_E z^2 \, dV$.

PART G
Mock Final Examination

28 problems

28 short-answer questions • 2 hours, closed book, one helpsheet

INSTRUCTIONS

Work the 28 questions below in one uninterrupted sitting of **two hours**. As in the real paper, only the final answer written in the box is marked, so check the requested form of each answer before writing it down. Unless a question says otherwise, give each answer as an integer or a fraction in lowest terms.

46. Evaluate $\displaystyle\int_0^1 x^3(\ln x)^2\,dx$.

Answer

47. Evaluate $\displaystyle\lim_{x\to 1}\frac{(x-1)\ln x}{x^2-3x+2+\ln x}$.

Answer

48. Consider the function f given by $f(x) = \displaystyle\int_{x+1}^{x^2+2} \frac{1}{t^3+1}\,dt$. Find $f'(1)$.

Answer

49. Let $P = (2, 1, -1)$ and $Q = (0, 2, 3)$. Find the point of intersection between the plane $x + 2y + z = 9$ and the line through P and Q.

Give the answer in the form (a, b, c), where a, b, c are real numbers.

Answer

50. Let S be the surface in xyz-space defined by the equation $2x^2 + y^2 - 3z = 3$. Find the point on S at which the tangent plane to S is parallel to the plane $8x - 4y - 6z = 7$.

Give the answer in the form (a, b, c), where a, b, c are real numbers.

Answer

51. Find the function $f(x, y)$ such that

$$\frac{\partial f}{\partial x} = 3x^2y + e^x, \qquad \frac{\partial f}{\partial y} = x^3 - 2\sin 2y,$$

and $f(0, 0) = 5$.

Give the answer as an expression in terms of x and y.

Answer

52. Let $u = x^2y$, let $v = v(x, y)$ be such that $\dfrac{\partial v}{\partial x} = -3x^2$ and $\dfrac{\partial v}{\partial y} = 4y$, let $w = x - 2y$, and let $z = uw + v$. Determine $\dfrac{\partial z}{\partial x}$ as a function of x, y.

Answer

53. Consider the function $f(x, y) = x^3 - xy^2$. Find the directional derivative of f at the point $(2, 1)$ in the direction of the vector $3\mathbf{i} - 4\mathbf{j}$.

Answer

54. Let $C(x, y, z)$, measured in parts per million, be the concentration of a gas at the point (x, y, z), where x, y, z are measured in metres. It is given that $\nabla C(x, y, z) = \langle 2x - z,\ 3y^2,\ 2 - x \rangle$. The position of a probe at time t seconds is $\mathbf{r}(t) = \langle 2t,\ t^2,\ t^3 - 1 \rangle$. Let A be the real number such that the rate of change of the concentration measured by the probe at time $t = 1$ is A parts per million per second. Find the value of A.

Answer

55. Determine all saddle points of the function $f(x, y) = 3x^2y + y^3 - 3y$.

Answer

56. Keep the notation of the previous question. Write down the system of three polynomial equations in the unknowns x, y and a parameter λ that you need to solve if you use the method of Lagrange multipliers to find the maximum value of $f(x, y) = 3x^2y + y^3 - 3y$ on the circle $x^2 + y^2 = 3$.

Answer

57. Keep the notation of the previous question. It is given that $(0, -1)$ is the only local maximum point of $f(x, y) = 3x^2y + y^3 - 3y$ in the xy-plane. Determine the maximum value of $f(x, y)$ on the closed disk of radius $\sqrt{3}$ centred at $(0, 0)$.

Answer

58. Let R be the region in the xy-plane bounded by the lines $x = 0$, $x = 3$, $y = 0$, $y = 2$. Evaluate $\iint_R \left(6xy^2 + 2x\right) dA$.

Answer

59. Consider the triangular region R in the xy-plane with vertices at $(0, 1)$, $(3, 1)$, and $(1, 4)$. Compute $\iint_R x \, dA$.

Answer

60. Let R be the region in the xy-plane bounded by the curve $y = x^2 - 2$ and the line $y = x$. Find the value of $\iint_R y \, dA$.

Answer

61. Find the exact value of the iterated integral $\displaystyle\int_0^4 \int_{\sqrt{x}}^2 \frac{1}{\sqrt{y^3+1}}\, dy\, dx$.

Answer

62. Let $f(t)$ be a function such that $\displaystyle\int_0^{12} f(t)\, dt = 30$. Let R be the region in the xy-plane bounded by the lines $x = 0$, $x = 4$, $y = 0$, and $y = 6$. Find the value of $\displaystyle\iint_R f(3x)\, f(2y)\, dA$.

Answer

63. Consider the region R in the first quadrant of the xy-plane bounded by the curve $y = \sqrt{2x - x^2}$ and the x-axis. Find the value of $\iint_R \frac{x}{x^2 + y^2}\, dA$.

Give the answer in terms of π.

Answer

64. Compute $\frac{1}{\pi} \iint_R \left(2 + 3x^4 + 6x^2y^2 + 3y^4\right) dA$, where R is the closed disk of unit radius centred at the origin.

Answer

65. Let D be the parallelogram in the xy-plane bounded by the lines $y = x$, $y = x + 3$, $y = -2x$, $y = -2x + 6$. Compute $\iint_D e^y \, dy \, dx$.

Give the answer as an expression in terms of integer powers of e.

Answer

66. Let R be the region in the first quadrant of the xy-plane bounded by the curves $x^2 + y^2 = 1$, $x^2 + y^2 = 4$ and the lines $y = 3x$, $y = 3x + 1$. Compute $\iint_R (x + 3y) \left(x^2 + y^2\right)^2 \, dy \, dx$.

Answer

67. Consider the double integral $\iint_R x^2 e^{xy}\, dA$, where R is the region in the first quadrant of the xy-plane bounded by the curves $xy = 1$, $xy = 3$ and the lines $y = x$, $y = 2x$. Use the change of variables $u = xy$ and $v = y/x$ to rewrite the double integral as an iterated integral in the variables u, v with integration order $du\, dv$. *[You do not have to compute the integral.]*

Give the answer in the form $\int_a^b \int_c^d g(u,v)\, du\, dv$.

Answer

68. Compute $\int_{-1}^{1}\int_{-1}^{1}\int_{-1}^{1} \left(x^2 + yz^3\right) dz\, dy\, dx$.

Answer

69. Compute the integral $\int_0^1 \int_0^x \int_{x+y}^3 dz\, dy\, dx$.

Answer

70. Let R be the solid region in xyz-space bounded by the surface $x^2 + y^2 = 4$, the surface $x^2 + y^2 = 1$, the plane $z = 3$, and the plane $z = 0$. Compute $\iiint_R z\left(x^2 + y^2\right)\,dx\,dy\,dz$.

Give the answer in terms of π.

Answer

71. Consider the solid region E in xyz-space bounded by the cylinder $x^2 + y^2 = 4$, the horizontal plane $z = 1$, and the surface $z = 4 + y$. Evaluate $\frac{1}{\pi}\iiint_E \left(x^2 + y^2\right)\,dV$.

Answer

72. Let E be the ball consisting of the points (x, y, z) in xyz-space such that $x^2 + y^2 + z^2 \leq 4$. Evaluate $\iiint_E \sqrt{x^2 + y^2 + z^2}\, dV$.

Give the answer in terms of π.

Answer

73. Let E be the solid region in xyz-space consisting of the points (x, y, z) such that $\sqrt{\frac{x^2 + y^2}{3}} \leq z \leq 3$. Rewrite the triple integral

$$\iiint_E \frac{z^2}{1 + x^2 + y^2 + z^2}\, dV$$

as an iterated integral in spherical coordinates with integration order $d\rho\, d\phi\, d\theta$. *[You do not have to compute the integral.]*

Answer

Answers

Exercise Set F

Consult these only after a complete attempt. Each answer is followed by the key step, so that a wrong answer can be traced to the line where it went wrong.

Part A

1. In general $\int_0^1 x^n \ln x\,dx = -\frac{1}{(n+1)^2}$ and $\int_0^1 x^n (\ln x)^2 dx = \frac{2}{(n+1)^3}$, the boundary terms vanishing by L'Hôpital. (a) $\frac{2}{27}$. (b) -4. (c) $\frac{1}{4}$ (parts with $u = \ln x$). (d) $\frac{2}{27}$ (parts twice, or $n!/a^{n+1}$).

2. (a) $\frac{2}{5}$: two applications of L'Hôpital give $\dfrac{1/x + 3/x^2}{2 - 3/x^2}$ at $x = 3$. (b) $\frac{9}{4}$. (c) e^6.

3. (a) $\dfrac{3x^2}{x^6+4} - \dfrac{2}{(2x-1)^2+4}$ at $x = 2$: $\frac{12}{68} - \frac{2}{13} = \frac{5}{221}$. (b) $2x\sqrt{1+x^6} - \sqrt{1+x^3}$ at $x = 2$: $4\sqrt{65} - 3$. (c) $F'(x) = 2x\left(x^2 - 4\right)e^{-x^2}$ changes from negative to positive at $x = 2$.

Part B

4. (a) $\mathbf{r} = (1 + 3t,\ -2 + 2t,\ 3 - 2t)$; $11 + 3t = 5$, $t = -2$: $(-5, -6, 7)$. (b) $\cos\theta = \frac{-3}{6}$, so $a = \frac{2}{3}$. (c) $\mathbf{n} = \langle 1, -1, 3\rangle \times \langle -1, 1, 1\rangle = \langle -4, -4, 0\rangle$: the plane is $x + y = 2$, and the distance is $\frac{5}{\sqrt{2}} = \frac{5\sqrt{2}}{2}$.

5. (a) $\langle 2x, 8y, -1\rangle = \lambda\langle 4, -8, -2\rangle$ gives $\lambda = \frac{1}{2}$, $x = 1$, $y = -\frac{1}{2}$, then $z = -3$: $\left(1, -\frac{1}{2}, -3\right)$. (b) $\nabla F = \langle 4, 2, 14\rangle$: $2x + y + 7z = 18$.

6. (a) $f = x^2e^y + y\cos x + y^2 + 3$; $f(\pi, 1) = \pi^2 e + 3$. (b) $g_{xy} = 2kxy + 3$ must equal $g_{yx} = 8xy + 3$, so $k = 4$ and $g = 2x^2y^2 + 3xy + y + C$.

7. $z_x = (2u + w)u_x - 3v_x + u\,w_x = 6x^2y + 8xy^2 - 2y^2 - 12x$.

8. $\nabla f(1, 2) = \langle 1, 4\rangle$. (a) $\frac{5-48}{13} = -\frac{43}{13}$. (b) $|\nabla f| = \sqrt{17}$.

9. $\mathbf{r}(1) = (1, 1, 1)$, $\nabla T = \langle 0, 2, 3\rangle$, $\mathbf{r}'(1) = \langle 2, 2, 3\rangle$: 13 °C/s.

Part C

10. $f_x = 2y(x - 1)$, $f_y = x^2 - 2x + 4y - 15$; $D = 8y - (2x - 2)^2$. Saddle points at $(5, 0)$ and $(-3, 0)$ ($D = -64$); local minimum -32 at $(1, 4)$ ($D = 32$, $f_{xx} = 8$).

11. Critical points $(0, 0)$ and $(2, 1)$; $D = 288xy - 144$. Saddle point at $(0, 0)$; local minimum -8 at $(2, 1)$. No local maximum.

12. (a) $\left(\frac{1}{2}, 0\right)$, a local minimum ($D = 8$), $f = -\frac{1}{4}$. (b) $2x - 1 = 2\lambda x$, $4y = 2\lambda y$, $x^2 + y^2 = 1$. (c) $y = 0$ gives $f(1, 0) = 0$, $f(-1, 0) = 2$; $\lambda = 2$ gives $x = -\frac{1}{2}$, $y = \pm\frac{\sqrt{3}}{2}$, $f = \frac{9}{4}$. Maximum $\frac{9}{4}$, minimum $-\frac{1}{4}$.

13. (a) $y^2 = 2x^2$, so $x = \pm 1$: maximum 2 at $(1, \pm\sqrt{2})$, minimum -2 at $(-1, \pm\sqrt{2})$. (b) $(4, -2, 4)$, at distance 6.

14. $y = 2x$, $z = 3x$, so $(x, y, z) = (2, 4, 6)$ and the maximum is $2 \cdot 16 \cdot 216 = 6912$.

Part D

15. (a) 10. (b) 1 (integrate in y first). (c) $2\ln 2 - 1$ (integrate in x first).

16. (a) $\left(\frac{1}{2}\cdot 12\right)\left(\frac{1}{4}\cdot 12\right) = 18$. (b) $5 + 5 = 10$.

17. $\displaystyle\int_1^5 \int_{(y+1)/2}^{(11-y)/2} y\,dx\,dy = \frac{56}{3}$ (area 8 times $\bar{y} = \frac{7}{3}$).

18. The curves meet at $x = -1$ and $x = 3$. (a) $\frac{32}{3}$. (b) $\frac{32}{3}$, so $\bar{x} = 1$.

19. $\displaystyle\int_0^1 \int_{y^2}^{2-y} y\,dx\,dy = \frac{5}{12}$.

20. (a) $\int_0^1 y^2\sqrt{1+y^3}\,dy = \frac{2}{9}\left(2\sqrt{2} - 1\right)$. (b) $\int_0^2 2xe^{x^2}dx = e^4 - 1$. (c) $\int_0^{\pi/2}\cos y\,dy = 1$. (d) $\int_0^2 \frac{y^3}{y^4+1}dy = \frac{1}{4}\ln 17$.

21. R is the triangle with vertices $(0, 0)$, $(2, 0)$, $(1, 1)$: $\displaystyle\int_0^1 \int_y^{2-y} xy\,dx\,dy = \frac{1}{3}$.

22. $\displaystyle\int_{-1}^1 \int_{x^2}^1 (2x + y + 1)\,dy\,dx = \frac{32}{15}$ (the term $2x$ integrates to 0 by symmetry).

Part E

23. (a) $\pi\left(1 - e^{-4}\right)$. (b) $\frac{31\pi}{5}$. (c) $\int_{\pi/4}^{\pi/3}\sin\theta\,d\theta\int_1^3 r^2 dr = \frac{13}{3}\left(\sqrt{2} - 1\right)$.

24. (a) $r = 4\cos\theta$, $0 \le \theta \le \frac{\pi}{2}$: $\int_0^{\pi/2} 4\cos\theta\,d\theta = 4$. (b) $r = 2\sin\theta$, $0 \le \theta \le \pi$: $\frac{8}{3}\int_0^{\pi}\sin^4\theta\,d\theta = \pi$. The disk has area π and centroid $(0, 1)$, so $\iint y\,dA = \bar{y}\cdot\text{area} = \pi$.

25. (a) The integrand is $r^4 - 2r^2 + 4$: $\frac{20}{3}$. (b) The region is $0 \le \theta \le \frac{\pi}{4}$, $0 \le r \le 2\sqrt{2}$: $\frac{\pi}{4}\ln 3$.

26. $\frac{1}{2}\int_{-\pi/6}^{\pi/6} 4\cos^2 3\theta\,d\theta = \frac{\pi}{3}$.

27. $I^2 = \int_0^{\pi/2}\int_0^{\infty} e^{-4r^2} r\,dr\,d\theta = \frac{\pi}{16}$, so $I = \frac{\sqrt{\pi}}{4}$.

28. $u = x - 2y$, $v = 2x + y$, $\partial(u, v)/\partial(x, y) = 5$: $\frac{1}{5}\int_0^4 u\,du\int_1^6 e^v dv = \frac{8}{5}\left(e^6 - e\right)$.

29. $\partial(u, v)/\partial(x, y) = 2v$ and $y^2 = uv$: $\int_1^4\int_1^2 \frac{u}{2}\,du\,dv = \frac{9}{4}$.

30. $u = x^2 - y^2$, $v = xy$, $\partial(u, v)/\partial(x, y) = 2\left(x^2 + y^2\right)$, so the integrand becomes $\frac{u}{2}$: $\int_1^4\int_1^3 \frac{u}{2}\,du\,dv = 6$.

31. $x = v\sqrt{u}$, $y = \sqrt{u}$, $|\partial(x,y)/\partial(u,v)| = \frac{1}{2}$: $\int_{1/3}^{1}\int_{1}^{4} \frac{1}{2}\sqrt{u}\,v^2\cos u\,du\,dv$.

32. $x = 3r\cos\theta$, $y = 2r\sin\theta$, Jacobian $6r$: $6\int_0^{2\pi}\int_0^1 r^3\,dr\,d\theta = 3\pi$.

33. Jacobian $\frac{1}{2}$; the triangle becomes $0 \le v \le 2$, $-v \le u \le v$: $\frac{1}{2}\int_0^2 v\left(e - e^{-1}\right)dv = e - \frac{1}{e}$.

Part F

34. (a) 22. (b) $\frac{13}{2}$. (c) 240: the first two terms are odd in x and in z respectively, and the box has volume 48.

35. (a) $\frac{2}{3}$. (b) $\frac{1}{6}$. (c) $\left(\frac{1}{2},\frac{1}{4},\frac{1}{2}\right)$, one quarter of the way to each intercept.

36. The shadow is the disk $r \le 2\sin\theta$: $\int_0^{\pi}\int_0^{2\sin\theta}\left(2r\sin\theta - r^2\right)r\,dr\,d\theta = \frac{4}{3}\int_0^{\pi}\sin^4\theta\,d\theta = \frac{\pi}{2}$.

37. $2\pi\int_1^2 r^4 dr\int_0^2 z\,dz = \frac{124\pi}{5}$.

38. $\frac{1}{\pi}\int_0^{2\pi}\int_0^2 r^3(4 - r\sin\theta)\,dr\,d\theta = 32$.

39. $2\pi\int_0^3\int_r^3 zr\,dz\,dr = \frac{81\pi}{4}$.

40. $2\pi\int_0^2 r^3(2-r)\,dr = \frac{16\pi}{5}$.

41. (a) $2\pi\int_1^2\rho^3 d\rho = \frac{15\pi}{2}$. (b) 4π. (c) By symmetry $\iiint x^2 = \frac{1}{3}\iiint\rho^2 = \frac{4\pi}{15}$.

42. The cone is $\phi = \frac{\pi}{6}$: $2\pi\left(1 - \frac{\sqrt{3}}{2}\right)\frac{8}{3} = \frac{8\pi}{3}\left(2 - \sqrt{3}\right)$.

43. (a) $\displaystyle\int_0^{2\pi}\int_0^{\pi/6}\int_0^{3\sec\phi}\frac{\rho^2\sin\phi}{1+\rho^2}\,d\rho\,d\phi\,d\theta$. (b) The ball is $\rho \le 2\cos\phi$, $0 \le \phi \le \frac{\pi}{2}$: $2\pi\int_0^{\pi/2} 4\cos^4\phi\sin\phi\,d\phi = \frac{8\pi}{5}$.

44. Slicing by horizontal disks of area πz: $M = \int_0^4 \pi z^2 dz = \frac{64\pi}{3}$ and $\bar{z} = \frac{1}{M}\int_0^4 \pi z^3 dz = 3$.

45. Jacobian 6, and $\iiint_{\text{unit ball}} w^2 = \frac{4\pi}{15}$: $\frac{8\pi}{5}$.

Part G

46. $\frac{1}{32}$. **47.** 2. **48.** $\frac{2}{28} - \frac{1}{9} = -\frac{5}{126}$.

49. $t = \frac{3}{2}$: $\left(-1, \frac{5}{2}, 5\right)$. **50.** $(1,-1,0)$.

51. $x^3y + e^x + \cos 2y + 3$. **52.** $3x^2y - 4xy^2 - 3x^2$.

53. $\nabla f = \langle 11, -4\rangle$: $\frac{49}{5}$. **54.** $\langle 4,3,0\rangle \cdot \langle 2,2,3\rangle = 14$.

55. $(1,0)$ and $(-1,0)$ (and $(0,1)$ is a local minimum, $(0,-1)$ a local maximum).

56. $6xy = 2\lambda x$, $3x^2 + 3y^2 - 3 = 2\lambda y$, $x^2 + y^2 = 3$.

57. Interior: $f(0,-1) = 2$. Boundary: $x = 0$ gives $f = 0$; $\lambda = 3y$ gives $y = \pm 1$, $x = \pm\sqrt{2}$, $f = 4y$. Maximum 4.

58. 90. **59.** $\int_1^4\int_{(y-1)/3}^{(11-2y)/3} x\,dx\,dy = 6$. **60.** $-\frac{9}{5}$.

61. $\int_0^2 \frac{y^2}{\sqrt{y^3+1}}dy = \frac{4}{3}$. **62.** $10\cdot 15 = 150$. **63.** $\int_0^{\pi/2} 2\cos^2\theta\,d\theta = \frac{\pi}{2}$.

64. 3. **65.** $u = y - x$, $v = y + 2x$, $y = \frac{2u+v}{3}$, Jacobian $\frac{1}{3}$: $\frac{3}{2}\left(e^2 - 1\right)^2$.

66. $u = x^2 + y^2$, $v = y - 3x$, $\partial(u,v)/\partial(x,y) = 2(x+3y)$: $\int_0^1\int_1^4 \frac{u^2}{2}\,du\,dv = \frac{21}{2}$.

67. $\displaystyle\int_1^2\int_1^3 \frac{u\,e^u}{2v^2}\,du\,dv$. **68.** $\frac{8}{3}$. **69.** 1.

70. $\frac{135\pi}{4}$. **71.** 24. **72.** 16π.

73. $\displaystyle\int_0^{2\pi}\int_0^{\pi/3}\int_0^{3\sec\phi}\frac{\rho^4\cos^2\phi\,\sin\phi}{1+\rho^2}\,d\rho\,d\phi\,d\theta$.

All problems in this set are original. They are modelled on the question types of the MA1521 sample final paper and of the Lecture Group 1 and Lecture Group 2 lesson notes for Weeks 8–11, and on the exercise sets of *Thomas' Calculus: Early Transcendentals*, 15th edition (Hass, Heil, Weir), Chapters 13 and 14.